

Teaching Exam

Subject – Computer Science

**Topic – Computer Organization &
Architecture**

Lecture No. – 06 ✓



By – Rahul Sir

Topics to be Covered

Topic



- 1:- Encoder ✓
- 2:- Types of Encoder ✓
- 3:- BCD , Hexadecimal , Octal all this to Binary
Conversion using Encoder ✓
- 4:- Priority Encoder ✓
- 5:- Advantage, Application , Disadvantage &
Conclusion On Encoder. ✓
- 6:- Decoder and its Various Types ✓

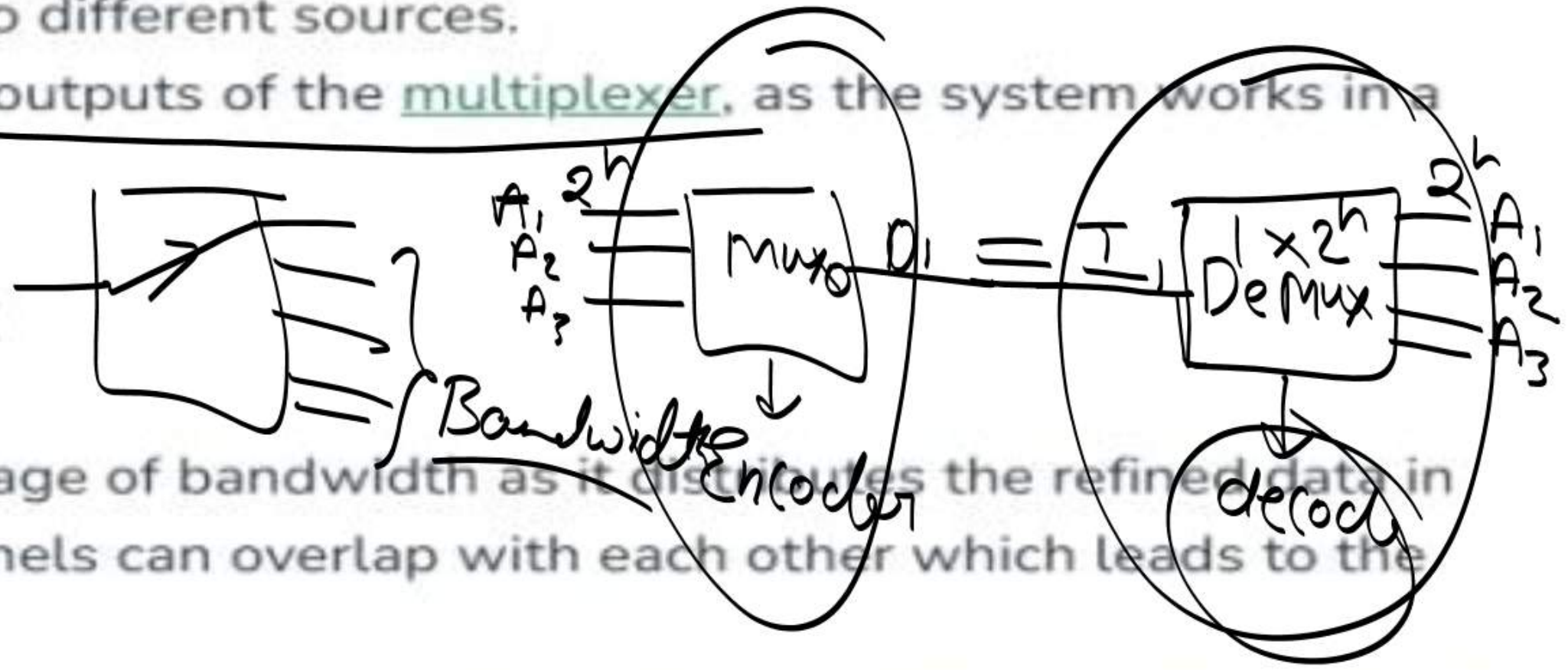


Advantages of the DEMUX

- The DEMUX increases the efficiency of the particular communication system as it takes data from a specific input source and distributes it to different sources.
- The DEMUX helps to separate the different signals from the mixed data sources. Then it distributes these data to different sources.
- DEMUX can decode the signal outputs of the multiplexer, as the system works in a reverse way of the MUX.

Disadvantages of the DEMUX

- The DEMUX may cause a wastage of bandwidth as it distributes the refined data in different channels. These channels can overlap with each other which leads to the loss of signal.
- The DEMUX may cause problems in the synchronization of signals. The data channels can overlap with each other which leads to the delay in the whole process.





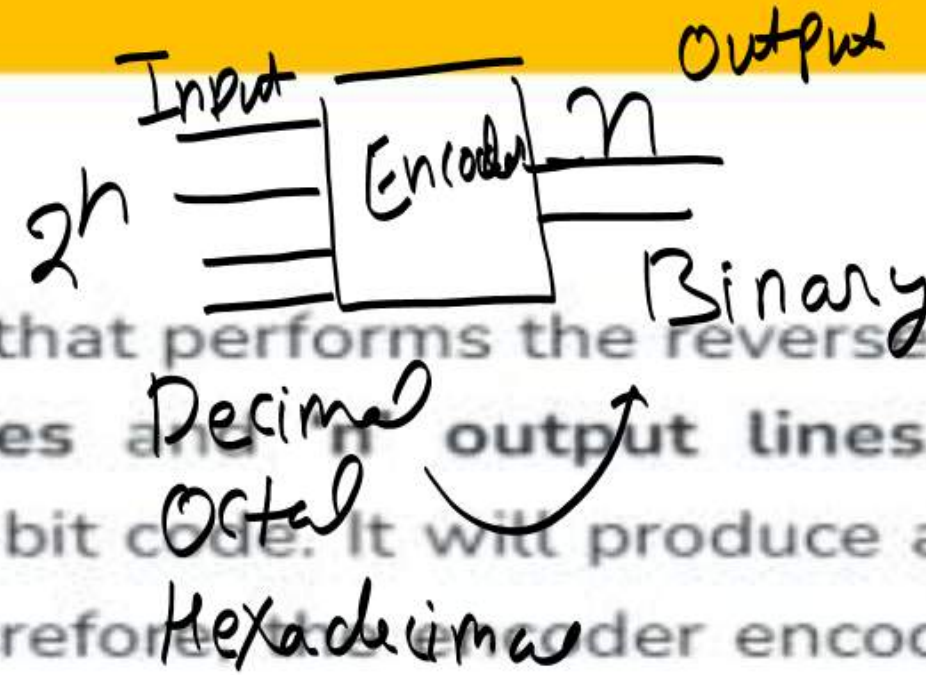
Applications of DEMUX

- DEMUXs are majorly used to design the ALU circuits and parallel data segments.
- The DEMUXs convert the output of the MUX into the actual input. On the receiver end, the DEMUX can be used to verify the original form of the data to carry out the entire communication process.
- DEMUX helps to save the output. The output generally saved to the ALU. The output will be saved in the registers and the various storage units of the system.
- Each DEMUX has a connection with multiple registers which helps to store the processed data into it.
- The counterpart of the DEMUX regulates the data signal at the output stage of the DEMUX operation. These data can be retrieved later to read out parallelly.
- In the audio or video system, the DEMUX distributes them in different channels. In broadcasting, the DEMUX separates the composite signal into an individual one.



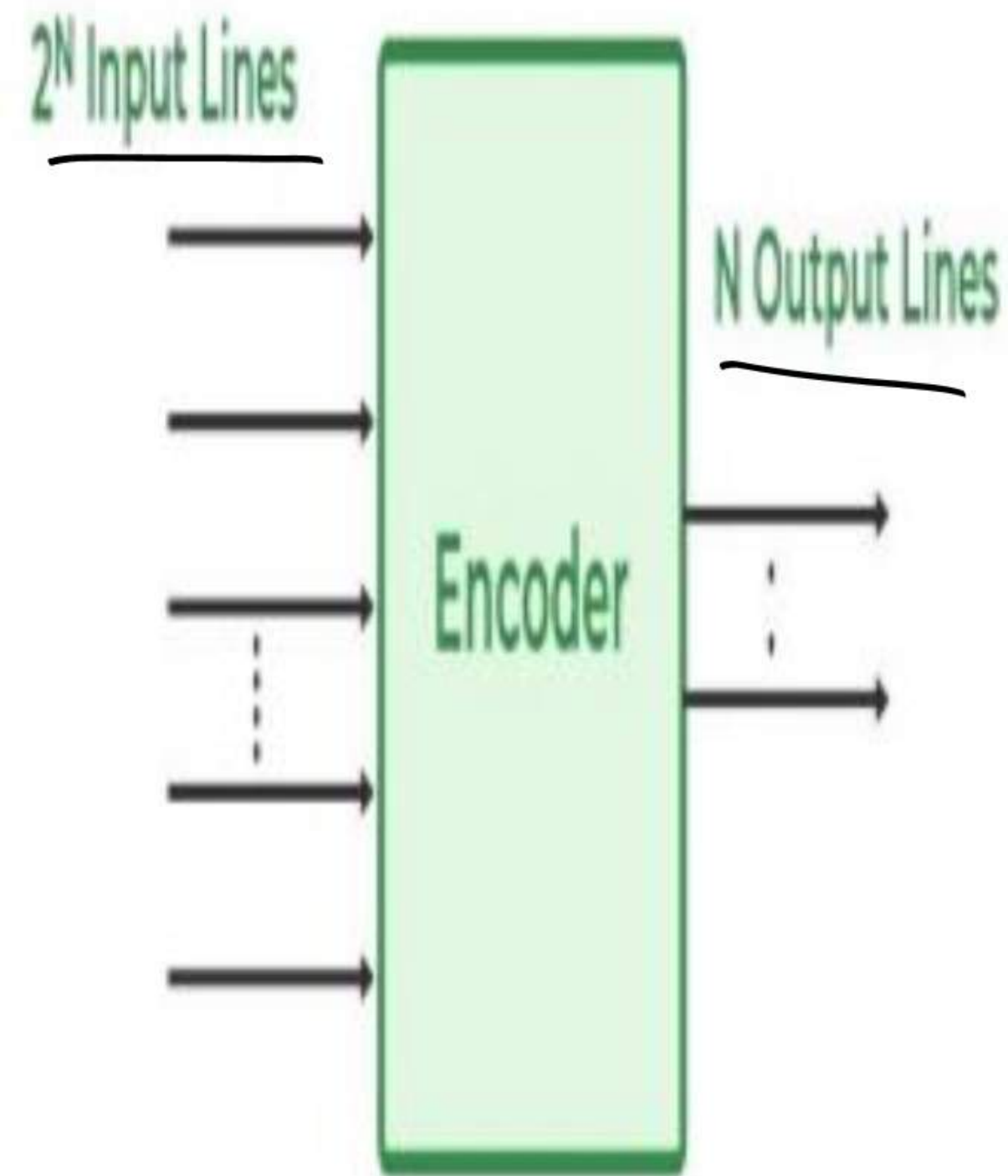
What is Encoder?

An Encoder is a **combinational circuit** that performs the reverse operation of a Decoder. It has a maximum of 2^n input lines and n output lines, hence it encodes the information from 2^n inputs into an n -bit code. It will produce a binary code equivalent to the input, which is active High. Therefore, the encoder encodes 2^n input lines with 'n' bits.



The basic principle of an encoder is to assign a unique binary code to each possible input. For example, a 2-to-4 line encoder has 2 input lines and 4 output lines and assigns a unique 4-bit binary code to each of the $2^2 = 4$ possible input combinations. The output of an encoder is usually active low, meaning that only one output is active (low) at any given time, and the remaining outputs are inactive (high). The active low output is selected based on the binary code assigned to the active input.

Note:- Decimal , Octal , Hexadecimal are all converted Binary output bit in Encoder



Types of Encoders

There are different types of Encoders which are mentioned below.

- 4 to 2 Encoder ✓
- Octal to Binary Encoder (8 to 3 Encoder)
- Decimal to BCD Encoder
- Priority Encoder

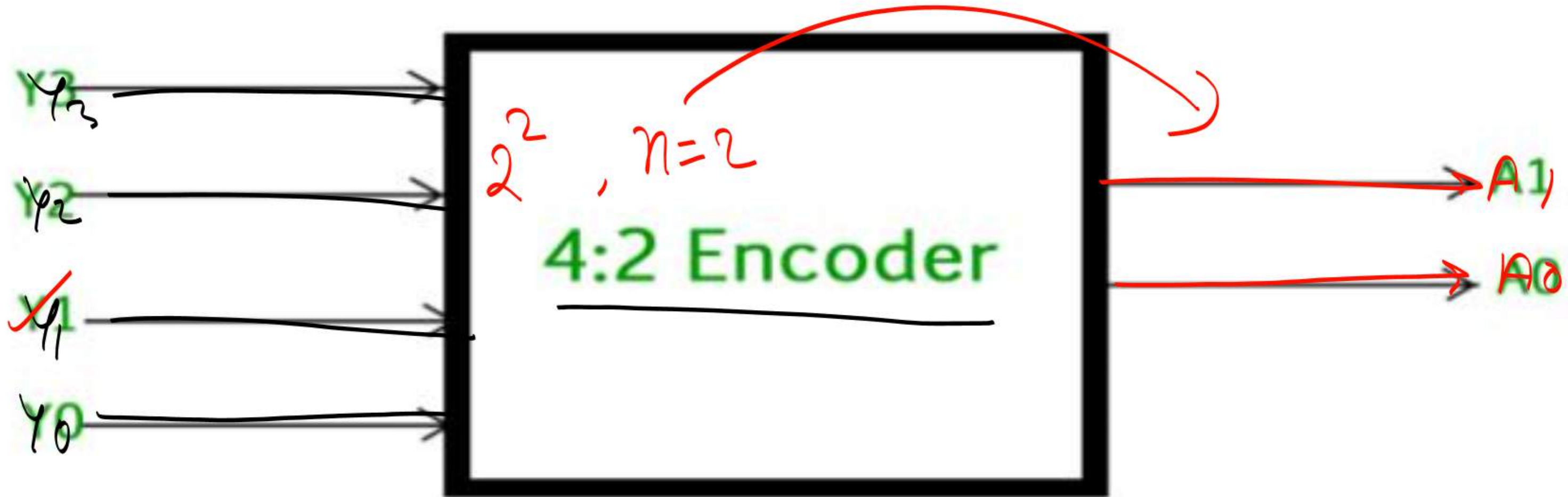


Topic :4 to 2 Encoder



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The 4 to 2 Encoder consists of four inputs Y_3 , Y_2 , Y_1 & Y_0 , and two outputs A_1 & A_0 . At any time, only one of these 4 inputs can be '1' in order to get the respective binary code at the output. The figure below shows the logic symbol of the 4 to 2 encoder.





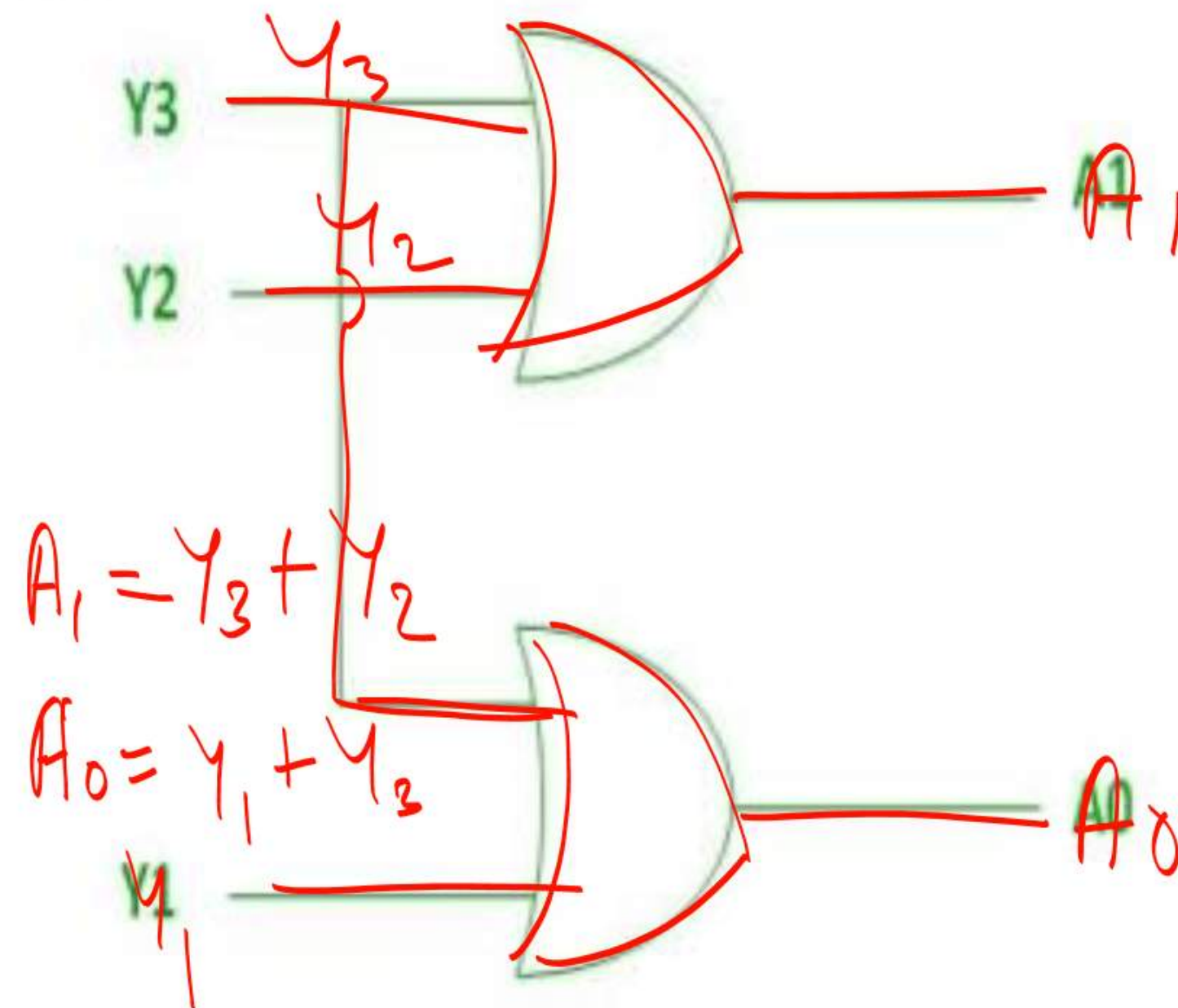
Topic :4 to 2 Encoder



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The Truth table of 4 to 2 encoders is as follows

INPUTS				OUTPUTS	
Y_3	Y_2	Y_1	Y_0	A_1	A_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1



Logical expression for A_1 and A_0 :

$$A_1 = Y_3 + Y_2$$

$$A_0 = Y_3 + Y_1$$

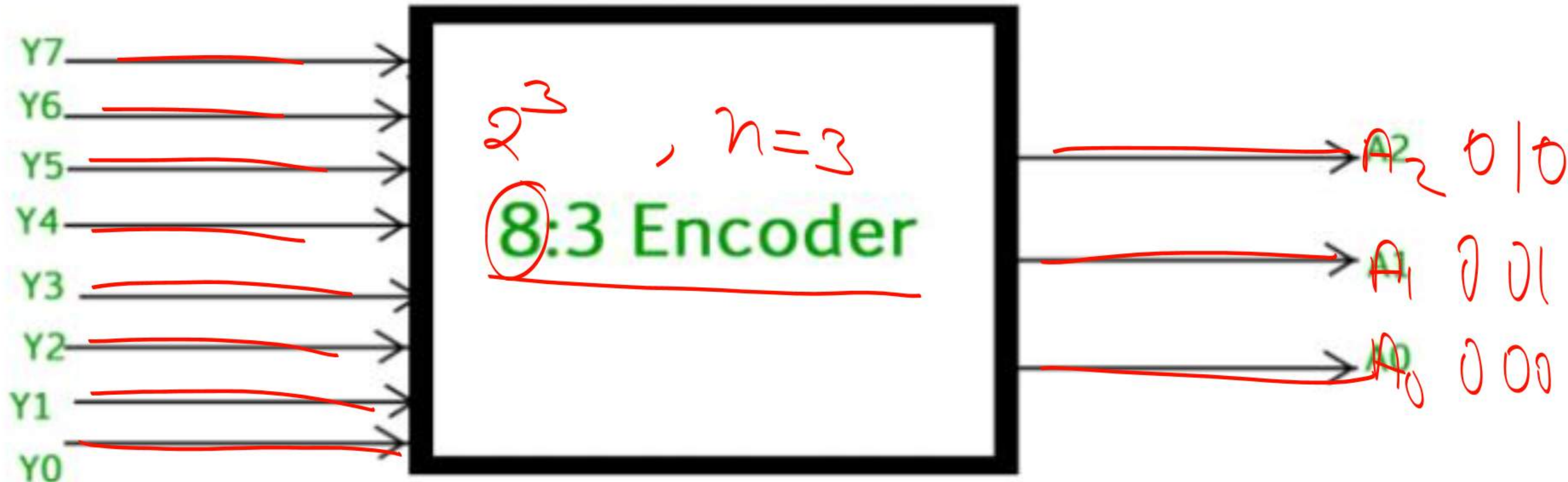


Topic : Octal to Binary Encoder



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The 8 to 3 Encoder or octal to Binary encoder consists of **8 inputs**: Y7 to Y0 and **3 outputs**: A2, A1 & A0. Each input line corresponds to each octal digit value and three outputs generate corresponding binary code. The figure below shows the logic symbol of octal to the binary encoder.





Topic : Octal to Binary Encoder



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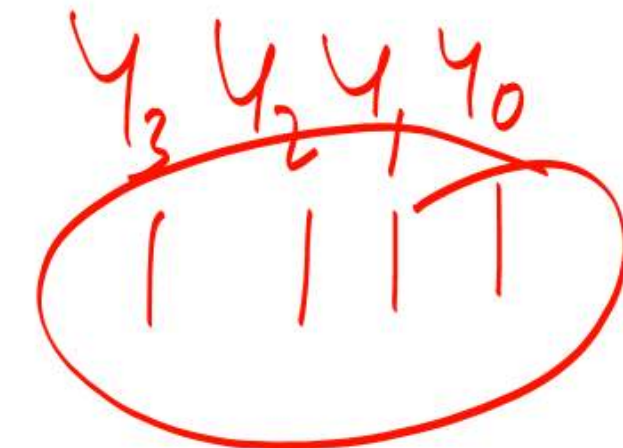
INPUTS								OUTPUTS		
Y7	Y6	Y5	Y4	Y3	Y2	Y1	Y0	A2	A1	A0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

Logical expression for A2, A1, and A0.

$$A_2 = Y_7 + Y_6 + Y_5 + Y_4$$

$$A_1 = Y_7 + Y_6 + Y_3 + Y_2$$

$$A_0 = Y_7 + Y_5 + Y_3 + Y_1$$

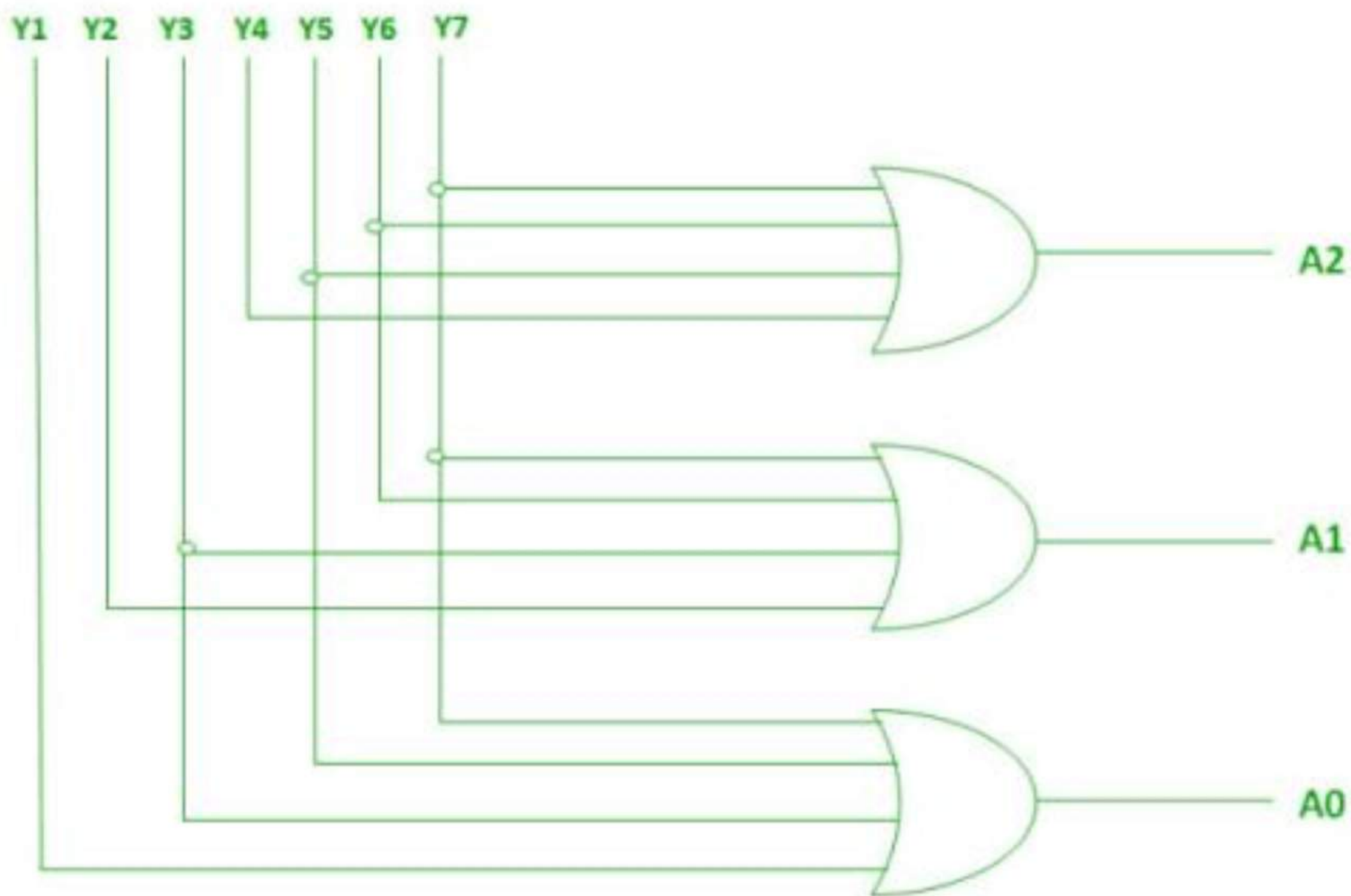




Topic : Octal to Binary Encoder



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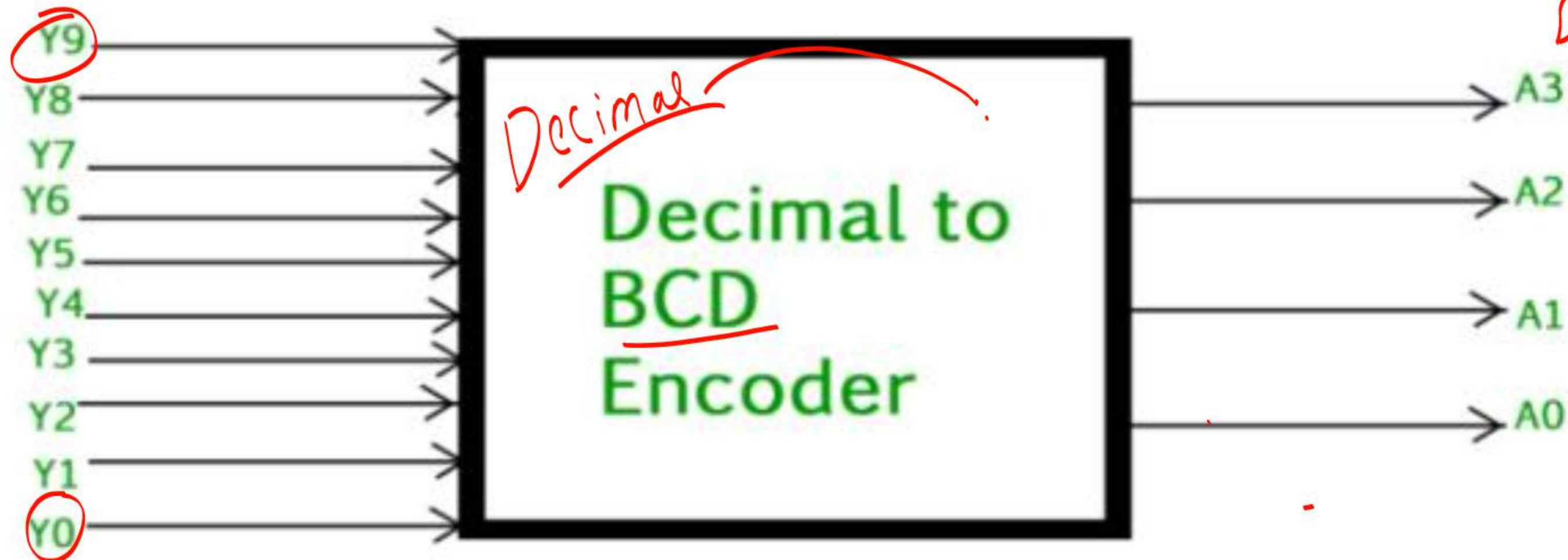


Topic : Decimal to Binary coded Encoder



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The decimal-to-binary encoder usually consists of **10 input lines** and **4 output lines**. Each input line corresponds to each decimal digit and 4 outputs correspond to the BCD code. This encoder accepts the decoded decimal data as an input and encodes it to the BCD output which is available on the output lines. The figure below shows the logic symbol of the decimal to BCD encoder :





Topic : Decimal to Binary coded Encoder



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INPUTS										OUTPUTS			
Y9	Y8	Y7	Y6	Y5	Y4	Y3	Y2	Y1	Y0	A3	A2	A1	A0
0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	1	0	0	0	0	1
0	0	0	0	0	0	0	1	0	0	0	0	1	0
0	0	0	0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	0	1	0	0	0	0	0	1	0	0
0	0	0	0	1	0	0	0	0	0	0	1	0	1
0	0	0	1	0	0	0	0	0	0	0	1	1	0
0	0	1	0	0	0	0	0	0	0	0	1	1	1
0	1	0	0	0	0	0	0	0	0	1	0	0	0
1	0	0	0	0	0	0	0	0	0	1	0	0	1



Topic : Decimal to Binary coded Encoder



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$$A_3 = \overset{\text{SOP}}{Y_9} + Y_8$$

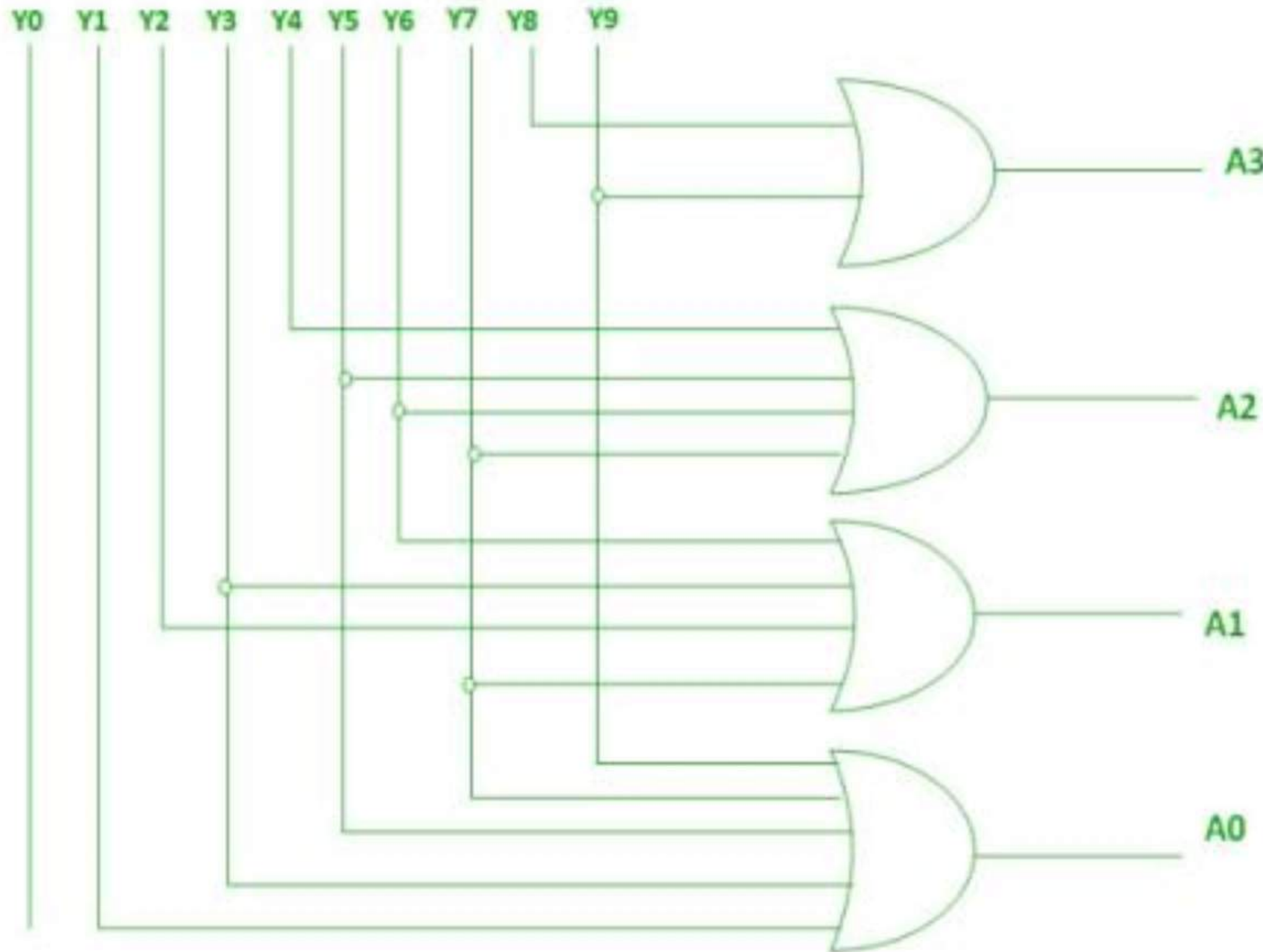
Logical expression for A3, A2, A1, and A0.

$$A_3 = Y_9 + Y_8$$

$$A_2 = Y_7 + Y_6 + Y_5 + Y_4$$

$$A_1 = Y_7 + Y_6 + Y_3 + Y_2$$

$$A_0 = Y_9 + Y_7 + Y_5 + Y_3 + Y_1$$





Topic : Priority Encoder



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A 4 to 2 priority encoder has **4 inputs**: Y3, Y2, Y1 & Y0, and **2 outputs**: A1 & A0. Here, the input, Y3 has the **highest priority**, whereas the input, Y0 has the **lowest priority**. In this case, even if more than one input is '1' at the same time, the output will be the (binary) code corresponding to the input, which is having **higher priority**. The truth table for the priority encoder is as follows.

INPUTS				OUTPUTS		
Y3	Y2	Y1	Y0	A1	A0	V
0	0	0	0	X	X	0
0	0	0	1	0	0	1
0	0	1	X	0	1	1
0	1	X	X	1	0	1
1	X	X	X	1	1	1

$$A_1 = Y_3 + Y_2$$
$$A_0 = Y_3 + Y_1$$

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Circuit ON or off in

Y3	Y2	Y1	Y0	A1	A0	V
X	X	X	1	0	0	1
X	X	1	0	0	1	1
X	1	0	0	1	0	1
1	0	0	0	1	1	1
0	0	0	0	X	X	0



Topic : Priority Encoder



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The logical expression for A1 is shown below.

A_1

$Y_1 Y_0$

$Y_3 Y_2$

	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$A_1 = Y_3 + Y_2$$

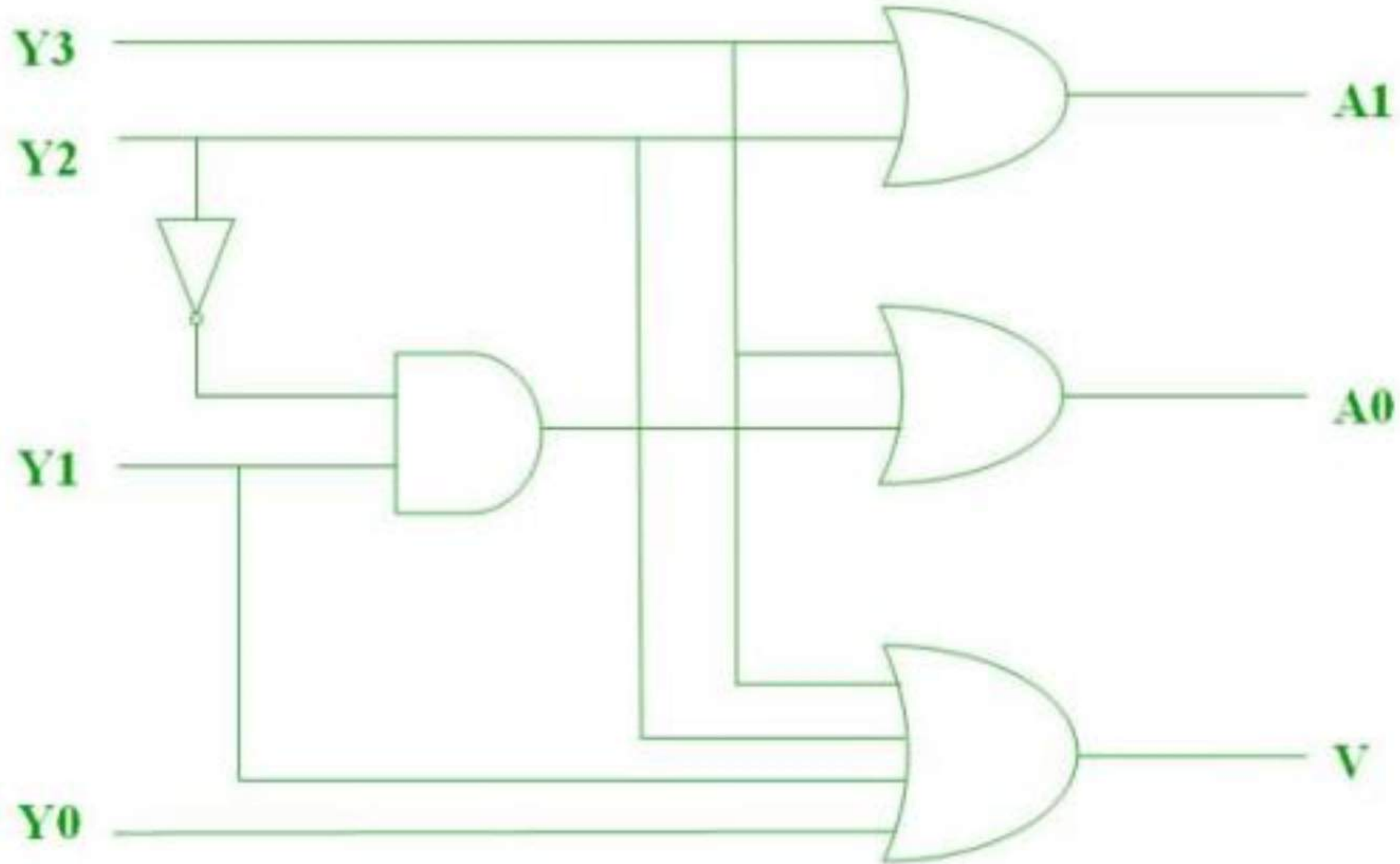
The Logical Expression for A0 is shown below.

$Y_1 Y_0$

$Y_3 Y_2$

	00	01	11	10
00	X	0	1	1
01	0	0	0	0
11	X	X	X	X
10	1	1	1	1

$$A_0 = Y_3 + Y_2' Y_1$$





Application of Encoders

- Encoders are very common electronic circuits used in all digital systems.
- Encoders are used to translate the decimal values to the binary in order to perform binary functions such as addition, subtraction, multiplication, etc.
- Other applications especially for Priority Encoders may include detecting interrupts in microprocessor applications.

Advantages of Encoders in Digital Logic

- **Reduction in the number of lines:** Encoders reduce the number of lines required to transmit information from multiple inputs to a single output, which can simplify the design of the system and reduce the cost of components.
- **Improved reliability:** By converting multiple inputs into a single serial code, encoders can reduce the possibility of errors in the transmission of information.
- **Improved performance:** Encoders can enhance the performance of a digital system by reducing the amount of time required to transmit information from multiple inputs to a single output.



Disadvantages of Encoders in Digital Logic

- **Increased complexity:** Encoders are typically more complex circuits compared to [multiplexers](#) , and require additional components to implement.
- **Limited to specific applications:** Encoders are only suitable for applications where a parallel set of inputs must be converted into a serial code.
- **Limited flexibility:** Encoders are limited in their flexibility, as they can only encode a fixed number of inputs into a fixed number of outputs.

Conclusion

Here we have learned about encoder on digital logic that ,an encoder is a device in digital electronics that takes multiple input signals and turns them into a smaller, coded output. Encoders are used in things like keyboards or communication systems to send or store information in a more compact way. We have learned about types of encoder, advantages disadvantages and applications of it. This helps make systems more efficient and less complex.



In digital electronics, a combinational logic circuit that converts an N-bit binary input code into M output channels in such a way that only one output channel is activated for each one of the possible combinations of inputs is known as a **decoder**.

In other words, a combinational logic circuit which converts N input lines into a maximum of 2^N output lines is called a **decoder**.

Therefore, a decoder is a combination logic circuit that is capable of identifying or detecting a particular code. The operation that a decoder performs is referred to as decoding. A general block diagram of a decoder is shown in Figure-1.

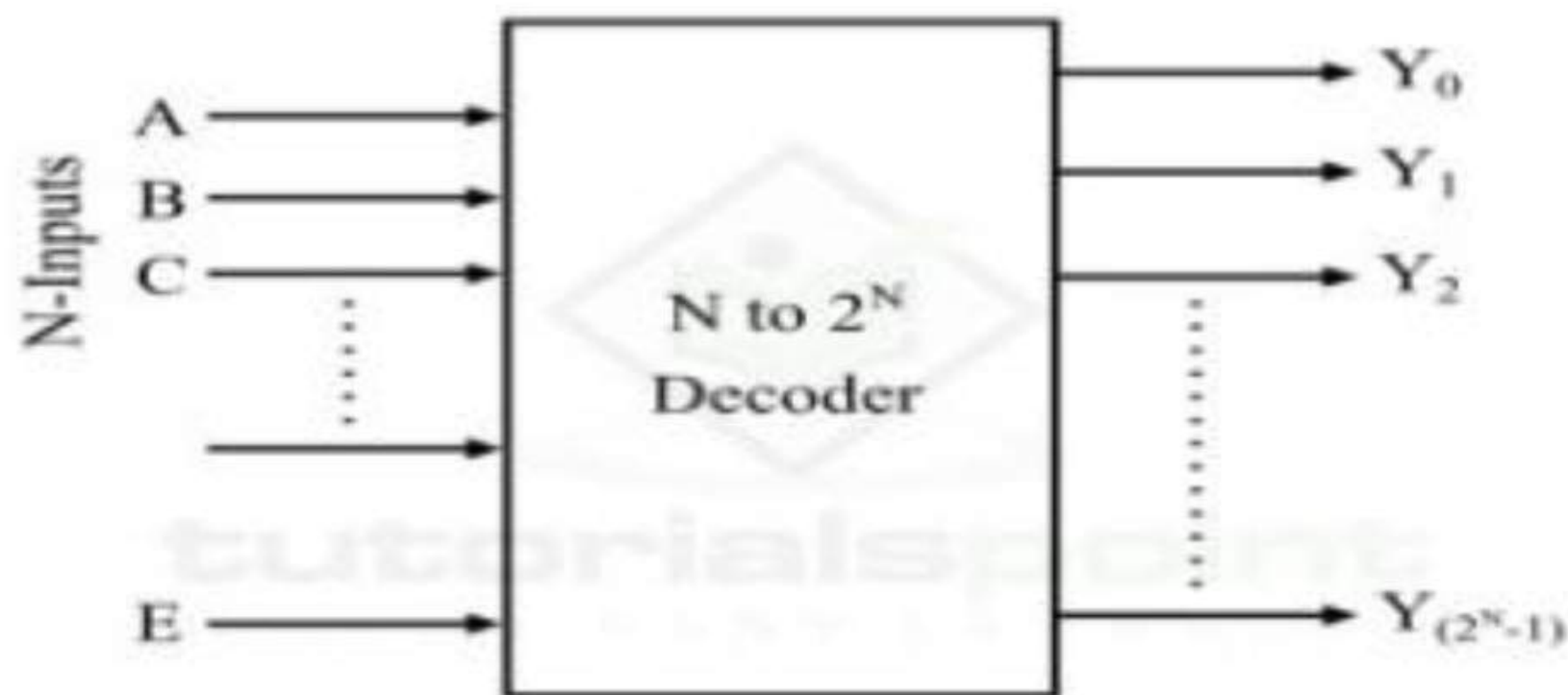


Figure 1 - Decoder

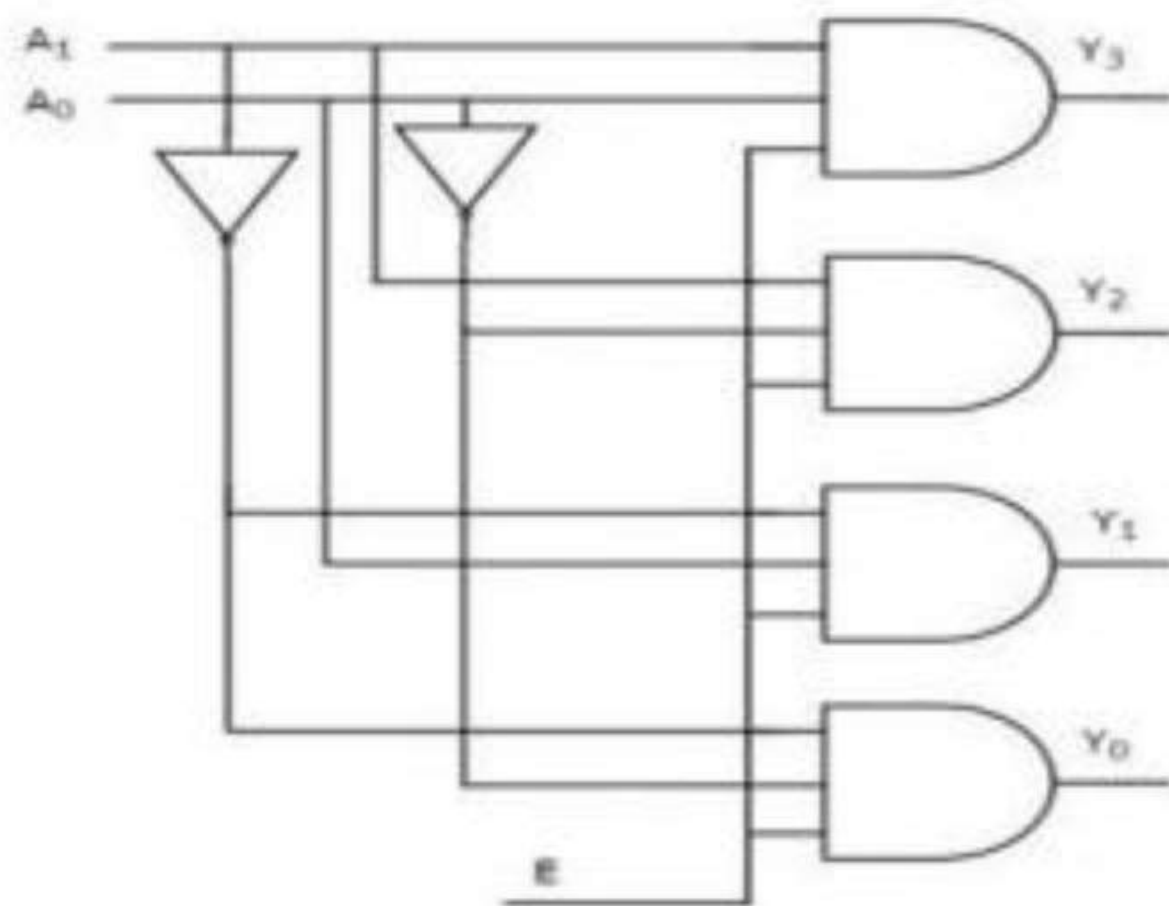
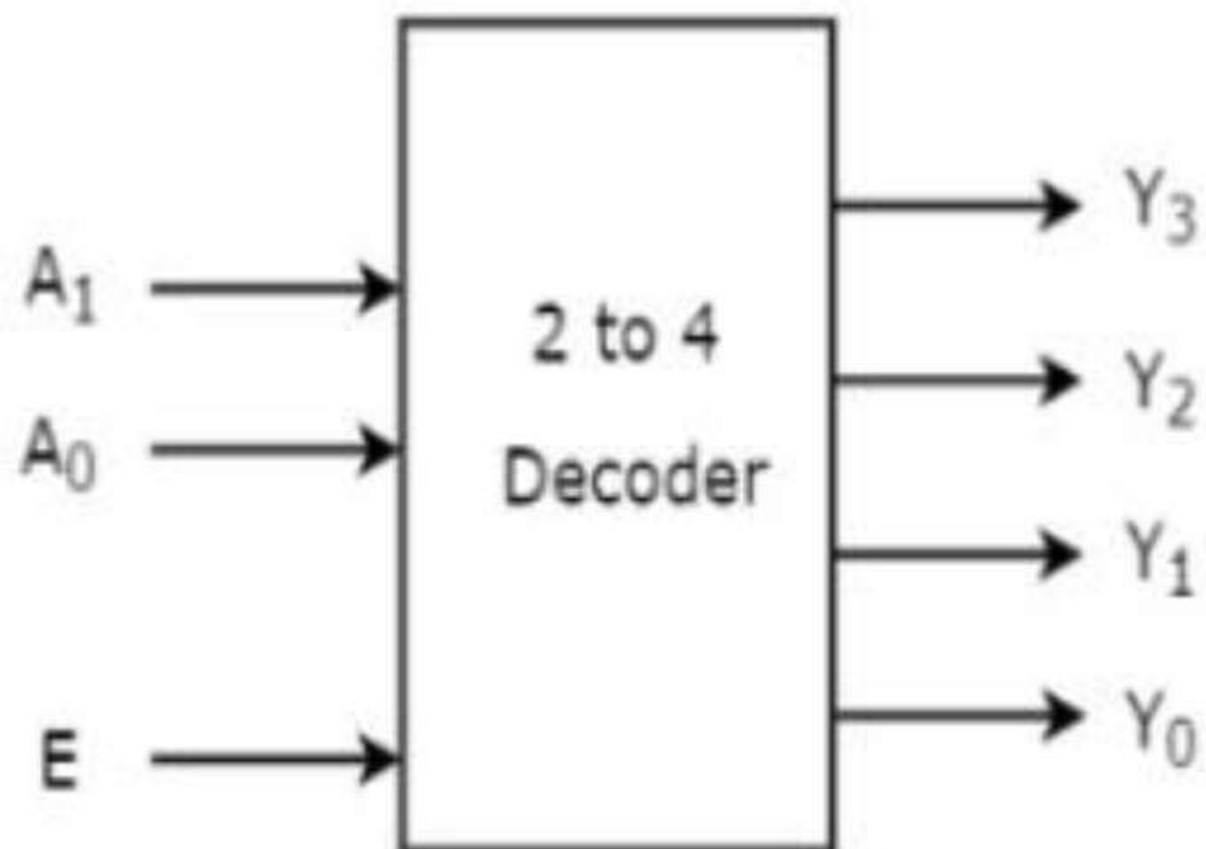


Here, the decoder has N input lines and M (2^N) output lines. In a decoder, each of the N input lines can be a 0 or a 1, hence the number of possible input combinations or codes be equal to 2^N . For each of these input combinations, only one of the M output lines will be active, and all other output lines will remain inactive.

Note:- Binary Bit at input are Converted in Octal, Decimal and Hexadecimal at output in Decoder



The 2 to 4 decoder is one that has 2 input lines and 4 (2^2) output lines. The functional block diagram of the 2 to 4 decoder is shown in Figure-2.



When this decoder is enabled with the help of enable input E , then its one of the four outputs will be active for each combination of inputs. The operation of this 2-line to 4-line decoder can be analyzed with the help of its truth table which is given below.



Inputs			Outputs			
E	A	B	Y_3	Y_2	Y_1	Y_0
0	X	X	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

Using this truth table, we can derive the Boolean expression for each output as follows –

$$Y_0 = E \cdot \bar{A} \cdot \bar{B}$$

$$Y_1 = E \cdot \bar{A} \cdot B$$

$$Y_2 = E \cdot A \cdot \bar{B}$$

$$Y_3 = E \cdot A \cdot B$$

THANK YOU